

British Airways Customer Reviews Dashboard

Internship Project Documentation

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Project: British Airways Customer Reviews Dashboard
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Internship Project Report

Chapter 1: Project Overview

1.1 Introduction

Customer reviews are an important source of feedback that help airlines understand passenger satisfaction and improve their services. This project develops an interactive Tableau dashboard to analyze British Airways customer reviews and present key insights through visualizations, KPIs, and filters for better decision-making.

1.2 Business Problem

British Airways collects customer reviews from passengers worldwide, but analyzing large volumes of raw data manually is difficult and time-consuming. An interactive dashboard helps organize this information into meaningful insights, enabling stakeholders to quickly identify service strengths, weaknesses, and customer satisfaction trends.

1.3 Project Objectives

The objective of this project is to design an interactive Tableau dashboard that provides a clear overview of customer feedback and service performance. The dashboard analyzes customer ratings based on traveller type, seat type, country, aircraft, recommendation status, and monthly trends to support data-driven business decisions.

1.4 Business Impact

The dashboard enables management to monitor customer satisfaction, compare service performance, and identify areas that require improvement. By converting customer review data into visual insights, it supports faster decision-making and helps improve the overall passenger experience.

1.5 Scope of the Project

This project focuses on analyzing historical British Airways customer review data using Tableau Public. It includes KPI analysis, interactive visualizations, dashboard filters, and performance comparisons, while advanced techniques such as sentiment analysis and predictive analytics are considered outside the current project scope.

Chapter 2: Dataset Understanding

2.1 Data Source

The dataset used in this project consists of customer reviews for British Airways collected from publicly available airline review platforms. It contains passenger feedback related to service quality, travel experience, and overall satisfaction, making it suitable for customer experience analysis.

2.2 Dataset Description

The dataset includes information such as customer ratings, traveller type, seat type, aircraft, country, recommendation status, travel date, and various service ratings. These attributes provide a comprehensive view of customer experiences and help identify trends and performance across different categories.

2.3 Data Dictionary

The table below summarizes the key attributes used in the dashboard:

Attribute	Description
Rating	Overall rating given by the customer.
Place	Country from which the review was submitted.
Aircraft	Aircraft used during the journey.
Traveller Type	Category of traveller (Business, Family, Solo, etc.).
Seat Type	Class of travel (Economy, Business, First, etc.).
Date	Date when the review was posted.
Cabin Staff Service	Rating for cabin crew service.
Seat Comfort	Rating for seating comfort.
Food & Beverages	Rating for onboard food and beverages.
Entertainment	Rating for in-flight entertainment.
Ground Service	Rating for airport and ground services.
Value for Money	Customer perception of overall value received.
Recommended	Indicates whether the customer recommends British Airways.

Chapter 3: Technology Stack

The primary tool used in this project was **Tableau Public**, which enabled the creation of an interactive and visually appealing dashboard. Tableau's features, such as KPI cards, charts, maps, filters, dashboard actions, and tooltips, were utilized to transform raw customer review data into meaningful business insights. Additionally, **Microsoft Excel** was used for basic data inspection and verification before importing the dataset into Tableau.

Technology Stack:

- **Tableau Public** – Dashboard development and data visualization
- **Microsoft Excel** – Data inspection and basic preprocessing

Chapter 4: Data Preparation

Before creating the dashboard, the dataset was examined to ensure its quality and consistency. The data was checked for **missing values, duplicate records, and formatting inconsistencies**. A few **null values were identified in the *Traveller Type* column**, while no significant duplicate records were found. Since the missing values represented only a small portion of the dataset, they were retained and did not have a noticeable impact on the overall analysis.

After validating the data, the appropriate data types for dates, ratings, and categorical fields were verified. The cleaned dataset was then imported into **Tableau Public**, where filters, calculated fields, and aggregations were applied to build accurate and interactive visualizations.

Chapter 5: Dashboard Analysis

5.1 Dashboard Overview

The complete interactive Tableau dashboard has been developed and is submitted as a separate Tableau workbook. It contains KPI cards, geographical analysis, trend analysis, aircraft analysis, traveller type analysis, seat type analysis, recommendation analysis, and interactive filters for dynamic exploration.

5.2 Dashboard Components & Interactivity

The dashboard consists of eight KPI cards that summarize important performance metrics, including **Average Rating, Total Reviews, Cabin Staff Service, Seat Comfort, Food & Beverages, Entertainment, Ground Service, and Value for Money**. It also includes multiple visualizations such as the **Average Rating by Country map, Monthly Trend, Traveller Type Analysis, Seat Type Analysis, Top 10 Aircraft Analysis, and Recommendation Rate**.

To improve user experience, interactive filters for **Country, Traveller Type, Seat Type, and Month** were added, enabling users to customize the dashboard according to their requirements. Dashboard actions and customized tooltips were also implemented so that selecting values in one visualization automatically updates the remaining charts, providing a seamless and interactive analytical experience.

5.3 Challenges Faced During Dashboard Development

Several challenges were encountered during the dashboard development process. Designing a balanced layout, maintaining consistent spacing between visualizations, creating professional KPI cards, and configuring dashboard actions required multiple design iterations. Ensuring that all charts remained visually aligned while preserving dashboard readability was also an important aspect of the development process.

Another challenge was encountered while creating the geographical map. **Four review locations could not be displayed because Tableau was unable to recognize or geocode those place names.** These records were excluded only from the map visualization, while they continued to be included in all other charts and KPI calculations to maintain the accuracy of the overall analysis.

5.4 Summary

The dashboard was successfully developed using Tableau Public by combining KPIs, interactive charts, filters, and dashboard actions into a single analytical interface. It provides a clear and interactive representation of British Airways customer reviews, enabling users to identify trends, evaluate service quality, and compare customer experiences across multiple categories.

Overall, the dashboard demonstrates how data visualization can simplify complex datasets and support informed business decisions through interactive and meaningful analysis.

Chapter 7: Key Business Insights

1. **Overall Customer Satisfaction:** Most passengers had a positive travel experience with British Airways, although there is still room to improve certain services and increase customer satisfaction.
2. **Average Rating by Country:** Customer ratings differ from one country to another, showing that passenger experiences are not the same across all regions. This helps the airline identify locations where service improvements may be needed.

3. **Monthly Trend Analysis:** Customer satisfaction changes throughout the year. Monitoring these monthly trends helps the airline understand when service quality improves or declines and plan improvements accordingly.
4. **Top 10 Aircraft Analysis:** Some aircraft received higher customer ratings than others. For example, the **Boeing 747-400** received one of the highest satisfaction scores, while the **Airbus A320** and **Boeing 777** were among the most frequently reviewed aircraft, showing they are widely used and provide a consistently good travel experience.
5. **Traveller Type Analysis:** Different groups of passengers, such as business travellers, solo travellers, couples, and families, rated their experiences differently. Understanding these differences helps British Airways better meet the needs of each type of traveller.
6. **Seat Type Analysis:** Premium travel classes such as **First Class** and **Business Class** generally received higher ratings than Economy Class, indicating that passengers value the additional comfort and services provided in premium cabins.
7. **Service Quality Analysis:** Customers rated different parts of their journey, including cabin staff service, seat comfort, food & beverages, entertainment, ground service, and value for money. These ratings help identify which services perform well and which areas require further improvement.
8. **Customer Recommendation Rate:** The recommendation analysis shows whether passengers would recommend British Airways to others. A higher recommendation rate reflects greater customer trust and satisfaction with the airline.
9. **Interactive Dashboard Features:** The dashboard includes filters for **Country**, **Traveller Type**, **Seat Type**, and **Month**, allowing users to explore customer reviews from different perspectives and gain insights based on their specific interests.

Chapter 8: Conclusion

This project successfully transformed British Airways customer review data into an interactive Tableau dashboard that makes customer feedback easy to understand and analyze. The dashboard provides valuable insights into customer satisfaction, service quality, and travel preferences through clear visualizations and interactive filters.

Overall, this project enhanced my practical skills in data visualization, dashboard design, and data analysis using Tableau. It also strengthened my understanding of how business intelligence tools can support data-driven decision-making and help organizations improve customer experience.